

PRACTICAL-03

AIM: Implementation of max-heap sort algorithm.

1.) MAX HEAP

PROGRAM:

```
#include <iostream>
```

```
using namespace std;
```

```
void maxheapify(int arr[], int size, int i)
```

```
{
```

```
    int largest = i;
```

```
    int l = 2 * i; int
```

```
    r = 2 * i + 1;
```

```
    if (l <= size && arr[l] > arr[largest])
```

```
    {
```

```
        largest = l;
```

```
    }
```

```
    if (r <= size && arr[r] > arr[largest])
```

```
    {
```

```
        largest = r;
```

```
    }
```

```
    if (largest != i)
```

```
    {
```

```
        swap(arr[largest], arr[i]);
```

```
        maxheapify(arr, size, largest);
```

```
    }
```

```
}  
  
void heapsort(int arr[], int size)  
{  
    for (int i = size / 2; i >= 1; i--)  
    {  
        maxheapify(arr, size, i);  
    }  
    for (int i = size; i > 1; i--)  
    {  
        swap(arr[1], arr[i]);  
        maxheapify(arr, i - 1, 1);  
    }  
}  
  
int main()  
{  
    int n;  
    cout << "Enter number of elements:  
"; cin >> n; int arr[n + 1]; cout <<  
"Enter elements: "; for (int i = 1; i <=  
n; i++)  
    {  
        cin >> arr[i];  
    }  
    heapsort(arr, n); cout  
<< "Sorted array: "; for  
(int i = 1; i <= n; i++)  
    {  
        cout << arr[i] << " ";  
    }  
    return 0;  
}
```

OUTPUT:

Output

```
Enter number of elements: 5
Enter elements: 5
9
7
6
1
Sorted array: 1 5 6 7 9

=== Code Execution Successful ===
```